INCH-POUND

MIL-PRF-39012/34B
09 September 2013
SUPERSEDING
MIL-C-39012/34A
11 January 1973

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY (SERIES TNC (UNCABLED) – RECEPTACLE, SOCKET, HERMETIC SEALED, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.

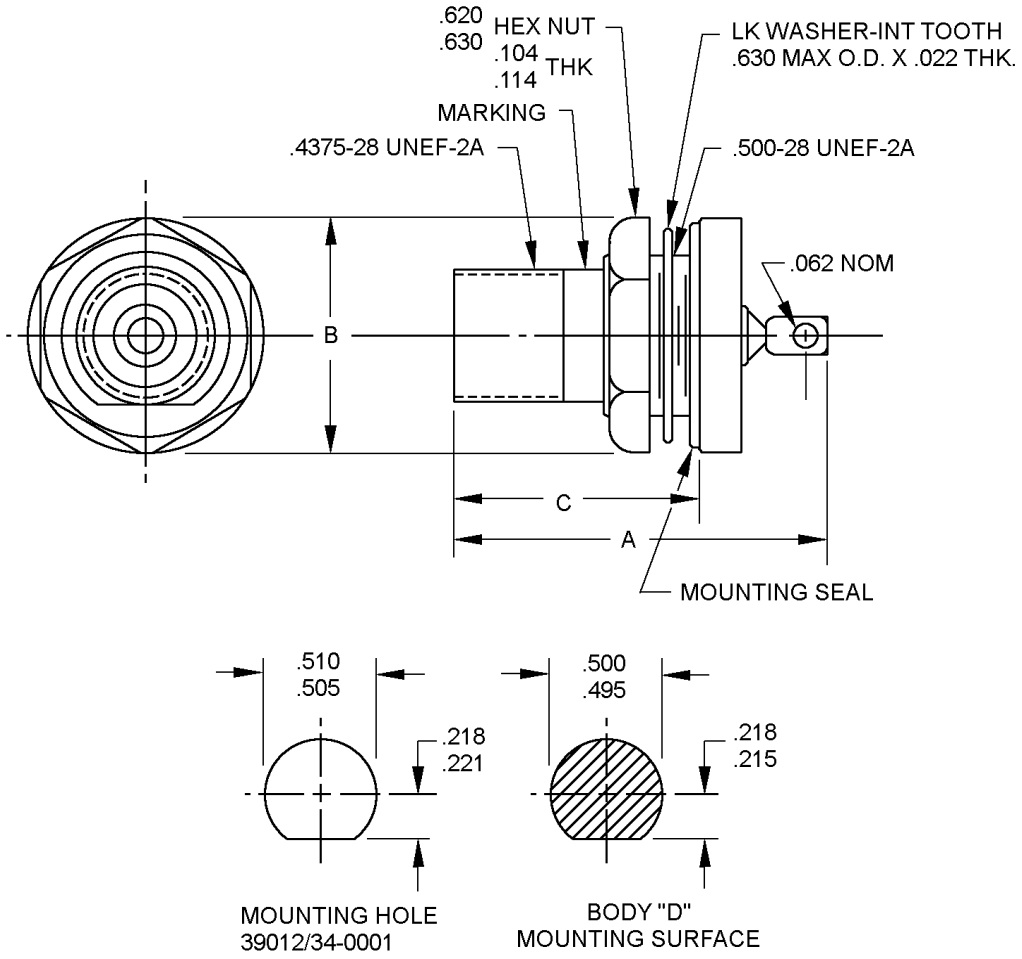
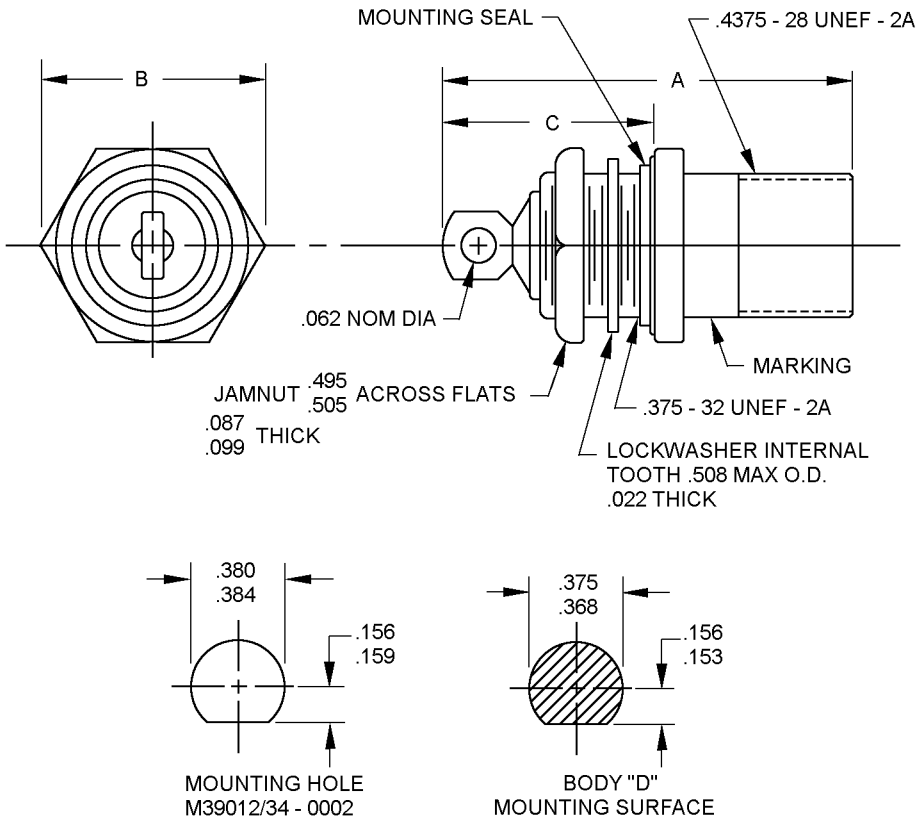


FIGURE 1. General configuration.

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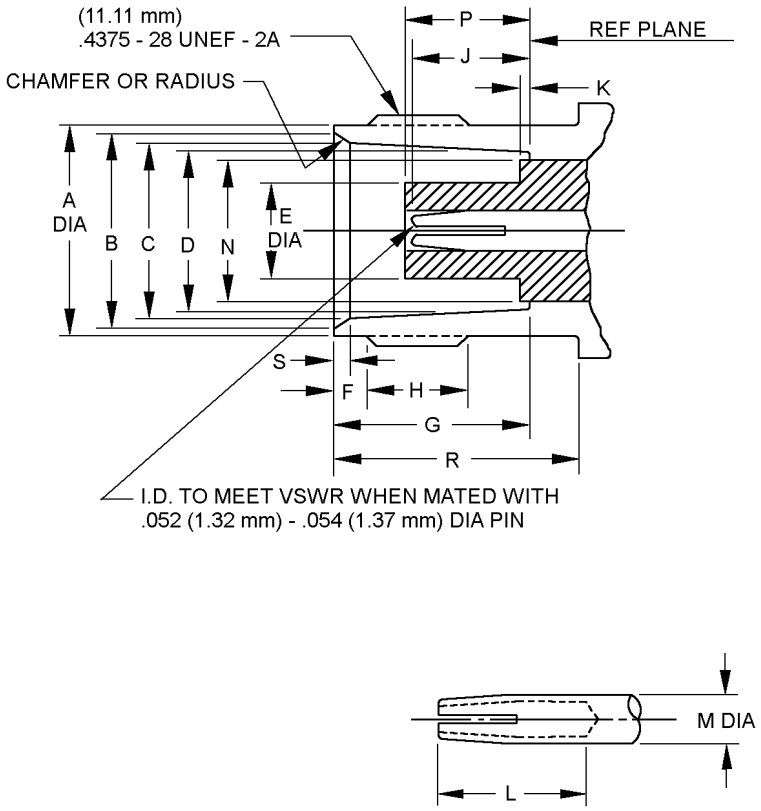


Inches	mm	Inches	mm	Inches	mm
.022	.56	.159	4.04	.4375	11.11
.062	1.57	.215	5.46	.495	12.57
.087	2.21	.218	5.54	.500	12.70
.099	2.51	.221	5.61	.505	12.83
.104	2.64	.368	9.35	.508	12.90
.114	2.90	.375	9.53	.510	12.95
.153	3.89	.380	9.65	.620	15.75
.156	3.96	.384	9.75	.630	16.00

NOTES:
1. Dimensions are in inches.
2. For dimensions A, B, and C see table I.
3. Dimension B is the largest overall diameter of the connector.
4. Metric equivalents (to the nearest .01 mm) are given for general information only.
5. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration – Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.378 (9.60)	.381 (9.68)
B	.345 (8.76)	.356 (9.04)
C	.327 (8.31)	.333 (8.46)
D	.319 (8.10)	.321 (8.15)
E		.186 (4.72)
F	.068 (1.73)	.088 (2.24)
G	.329 (8.36)	.333 (8.46)
H	.187 (4.75)	
J	.186 (4.72)	.206 (5.23)
K		.006 (.15)
L	.195 (4.95)	
M	.081 (2.06)	.087 (2.21)
N		.256 (6.50)
P	.188 (4.78)	.208 (5.28)
R	.415 (10.56)	
S	.015 (.38)	.030 (.76)

1. Metric equivalents (to the nearest .01 mm) are given for general information only.
2. All undimensioned pictorial configurations are for reference purposes only.
3. N dimension applies to that portion (If applicable of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension K.

FIGURE 2. Mating dimensions for female terminations.

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TABLE I. Dash number and overall dimensions.

Dash No.	Dim	Inches-millimeters 1/		Maximum panel thickness
		Minimum	Maximum	
0001	A		1.343 (34.11)	.125 (3.18)
	B	.684 (17.37)	.690 (17.53)	
	C	.769 (19.53)	.893 (22.68)	
0002	A		1.328 (33.73)	.250 (6.35)
	B	.630 (16.00)	.650 (16.51)	
	C	.626 (15.90)	.750 (19.05)	

1/ Millimeters are in parentheses.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	0001 0002	0001 0002

ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating: 500 volts rms maximum working voltage at sea level, 125 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force – Not applicable.
Torque – 2 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions:
Coupling torque – 4 to 6 inch-pounds.

Mating characteristics: See figure 2 for dimensions.
Center contact (female)
Oversize test pin - .057 diameter minimum (non-closed entry contacts only).
Insertion depth - .125 minimum.
Number of insertions – 1.

Insertion force test – Steel test pin dia .054 minimum.
Test pin finish – 16 microinches.
Insertion force – 2 lbs maximum.

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Withdrawal force test: Steel test pin dia .052 maximum.
Withdrawal force – 2 oz minimum.
Test pin finish – 16 microinches.

Hermetic seal: Leakage shall not exceed 1 x 10^{-5} cc/sec of tracer gas at atmospheric pressure.

Leakage (pressurized connectors): Connector mounted in mounting hole specified on figure 1 with mating end capped. Test applicable to mounting seal only. Air pressure – 30 psi. Duration: 30 seconds minimum.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum.

Center contact retention:
6 lbs minimum axial force.
4 inch-ounces radial torque minimum.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): Not applicable.

Connector durability: 500 cycles at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

Contact	Initial	After environment
Center contact	4.0	5.0
Outer contact	.2	Not applicable.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 1,500 volts rms minimum at sea level.

Vibration, high frequency: Method 204, test condition C, MIL-STD-202.

Group qualification: See table II.

Shock: Method 213 of MIL-STD-202, test condition I.

Temperature cycling: Method 102, test condition C, MIL-STD-202, except test high temperature shall be +200°C.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:
Voltage – 375 volts, minimum.
Altitude – 70,000 feet.

RF high potential withstanding voltage:
Voltage and frequency: 1,000 volts rms at 5 MHz.
Leakage current: Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

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RF leakage: Not applicable.

Insertion loss: Not applicable.

Part or Identifying Number (PIN): M39012/34 – (dash number from table I)

Changes from previous issues. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

MIL-STD-202

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AT, AV, EA, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99
NASA - NA

(Project 5935-2013-194)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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